EXAMPLE:

The zero of main scale matches with zero of vernier scale. 9th division of main scale matches with 10th division of vernier. No other match is present. After sliding, zero of vernier is found to be in between 3 & 4th divisions of main scale & 6th divison of vernier matches with xth division of main scale. The value of x/3 is

Solution: Refer to the figure, this is a standard Vernier with reading 3·6 mm.

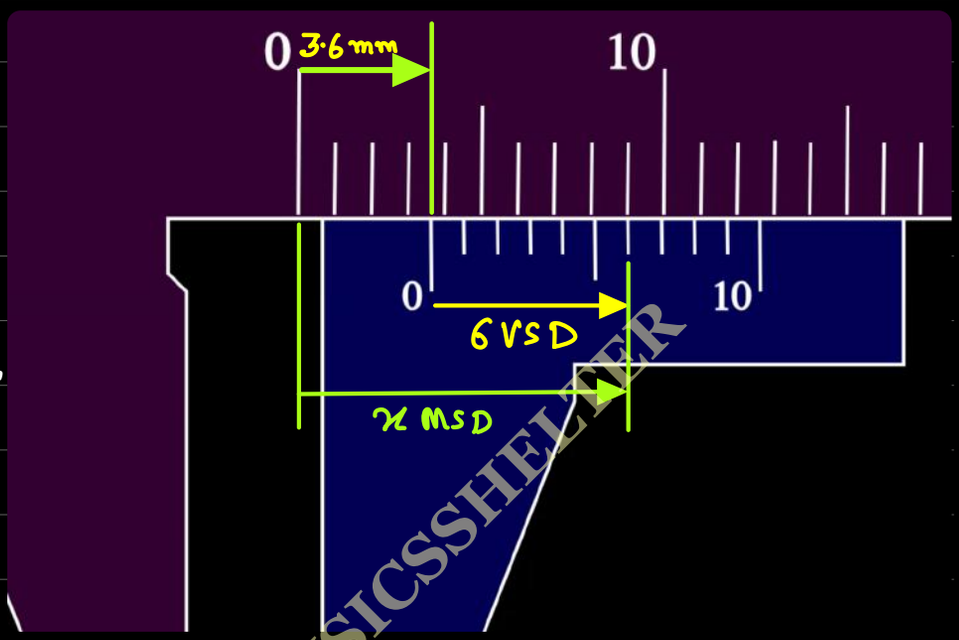
Thus:

$3 \cdot 6\text{mm} + 6\,\text{VSD} = x\,\text{MSD}$

$\Rightarrow x\,\text{MSD} = (3 \cdot 6 + 6 \times 0 \cdot 9)\text{mm}$

$\Rightarrow x\,\text{MSD} = 9\text{ mm}$

$\Rightarrow \frac{x}{3} = 3$ ans.



EXAMPLE:

In ordinary Vernier calipers, 10th division of the Vernier scale coincides with 9th division of the main scale. In a specially designed Vernier calipers the Vernier scale is so constructed that 10th division on it coincides with 11th division on the main scale. Each division on the main scale equals to 1 mm. The calipers have a zero error as shown in the figure-I. When the Vernier caliper is used to measure a length, the concerned portion of its scale is shown in figure-II.

Figure-I

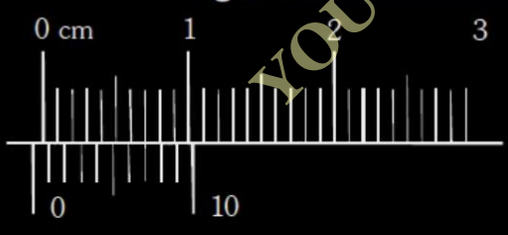


Figure-II



(A) Zero error in the calipers has magnitude 0.7 mm.

(B) The length being measured is 1.08 cm.

(C) The length being measured is 1.22 cm.

(D) Though the given Vernier scale does not follow the principle of Vernier scale, yet can be used satisfactorily.

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Solution: This is a standard Retrograde Vernier, For zero error we can either use algorithm. (Imagine -1 etching to the left of zero, Take MSR to the right of Vernier zero and subtract according to coinciding division) or use direct logic. Here it is a case of negative zero error. Let negative error be x mm. Now 7th VSD is coinciding with 7th MSD $\Rightarrow$ $x + 7\,MSD = 7\,VSD$

$\Rightarrow x = 7(VSD - MSD) = 7(1.1\,mm - 1\,mm)$

$\Rightarrow x = 0.7$ mm. $\Rightarrow$ Zero Error $= -0.7$ mm.

Now we shall find reading assuming no zero error. We can see that 5th Vernier mark coincides with 17 mm mark on Main Scale Now let reading be y $\Rightarrow$

$y + 5\,VSD = 17$ mm $\Rightarrow y = 17\,mm - 5.5$ mm

$\Rightarrow y = 11.5$ mm. Now for correct reading we need to subtract zero error.

$\therefore$ Correct reading : $11.5\,mm - (-0.7\,mm) = 12.2$ mm.

$\therefore$ Correct length is 12.2 mm obs. (A & C are correct).



Using Algorithm: Zero Error $= 0 - 7 \times 0.1 = -0.7$ mm.

Reading : $12 - 5 \times 0.1 = 11.5$ mm.

Correct Reading: $11.5 - (-0.7) = 12.2$ mm.

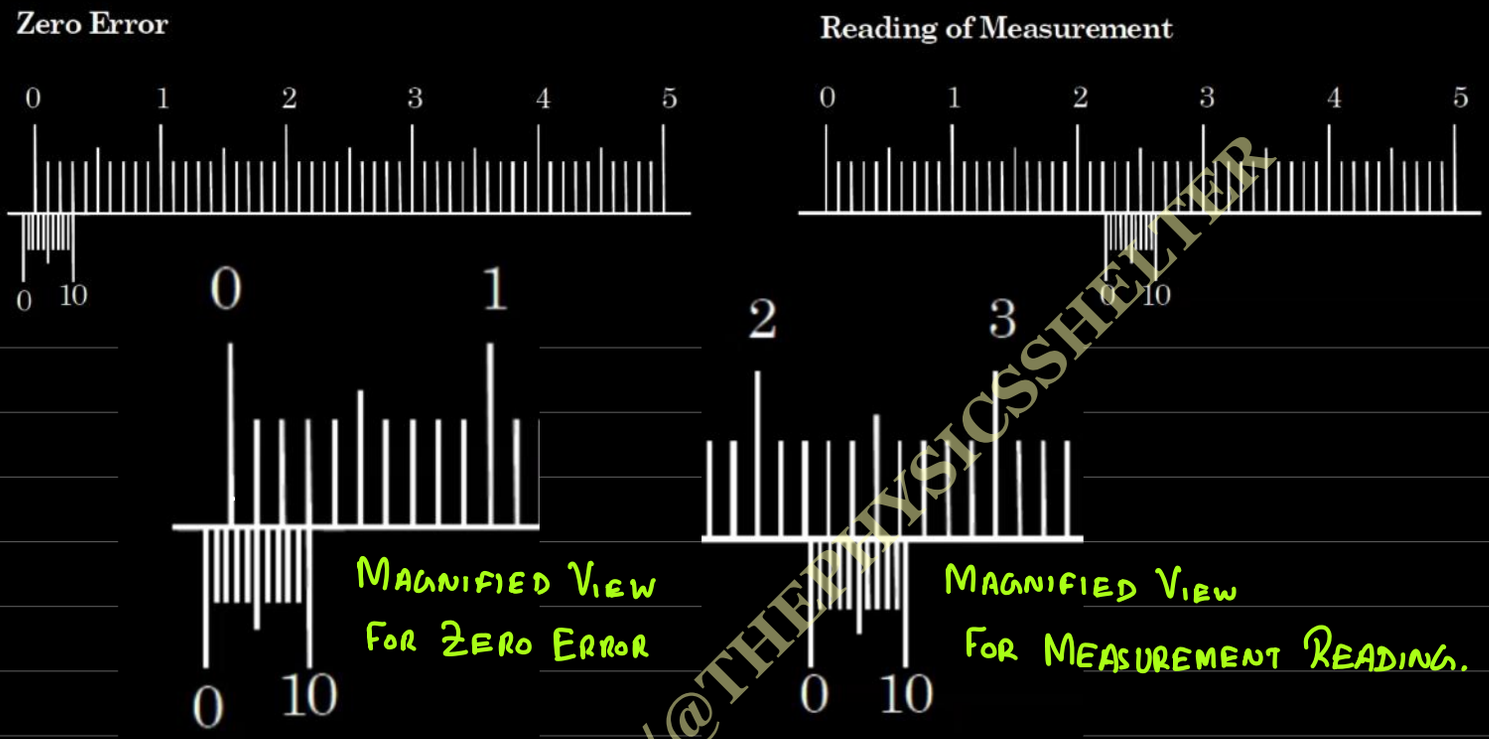
[In Algorithm, we take main scale reading to the right of Vernier zero and subtract according to coinciding division]]

Recall standard theory of Retrograde Vernier]

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EXAMPLE:

In a specially designed vernier calipers, 10 divisions of the vernier scale are equal to 4 divisions of the main scale. In the following figure, the zero error and a measurement are shown. In the zero error, the 5th and the 10th divisions of the vernier coincides with 1 mm and 3 mm mark on the main scale respectively and in the measurement, the 2nd and the 7th divisions of the vernier scale coincide with the 23 mm and 25 mm mark on the main scale respectively as shown in the figure. Report the length measured with due regards to the error.



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Solution: $10\,VSD = 4\,mm \Rightarrow 1\,VSD = 0.4\,mm$. Let negative zero error be x mm. Now 5th VSD coincided with 1st MSD

$\Rightarrow x + 1\,MSD = 5\,VSD \Rightarrow x = 5 \times 0.4 - 1 = 1\,mm.$

$\Rightarrow$ Zero Error $= -1$ mm.

Now Reading: 7th VSD is coinciding with 25 mm mark on mainscale

Let Reading be $y \Rightarrow y + 7\,VSD = 25\,mm \Rightarrow y = 25 - 7 \times 0.4$

$\Rightarrow y = 22.2$ mm.

$\Rightarrow$ Correct Reading $= y -$ Zero Error $= 22.2 - (-1)$ mm

$\Rightarrow$ Reading $= 23.2$ mm. Now least count of Vernier $=$ GCD (1mm, 0.4mm)

$\Rightarrow$ L.C. $= 0.2$ mm. $\Rightarrow$ Correct Reading : 23.2 ± 0.2 mm.

Note: Vernier Constant $= 1\,MSD - 1\,VSD$ (This may not always be equal to least count).

EXAMPLE:

Consider a pair of vernier calipers. Least count of main scale is 1mm. There are 20 equally spaced marks on vernier scale. When nothing is put between the jaws, zero of main scale lies between zero and first division of vernier scale and 15th division of vernier exactly coincide with one of the main scale division. When zero of vernier is matched with zero of main scale, 20th division of vernier exactly coincides with 25th division of main scale. To measure the side of a cuboid, it is kept between the jaws of vernier calipers, it is now observed that 18th division of vernier exactly coincides with 32nd division of main scale. Then

A) Least count of vernier scale is 0.50 mm

B) Zero error is -0.25 mm

C) Side of cuboid is 10.25 mm

D) Side of cuboid is 9.25 mm

Solution: Here zero error is negative. Also $20\,VSD = 25\,MSD$

$\Rightarrow 1\,VSD = 1.25\,mm$. Let negative zero error be x

$\Rightarrow x + n\,MSD = 15\,VSD$, Here n is unknown however we know that x is less than 1.25mm and greater than zero

Now $x = 15\,VSD - n\,MSD = 15 \times 1.25\,mm - n = 18.75 - n$

$\Rightarrow 18.75 - n < 1.25$ and $18.75 - n > 0$

$\Rightarrow n > 17.5$ and $n < 18.75 \Rightarrow n = 18$.

$\Rightarrow x = 15\,VSD - 18\,MSD = 18.75 - 18 = 0.75$

$\Rightarrow$ Zero Error $= -0.75\,mm$.

Now let side of cube be $y \Rightarrow y + 18\,VSD = 32\,mm$

$\Rightarrow y = 32\,mm - 18 \times 1.25 = 9.5\,mm$.

$\Rightarrow$ CORRECT READING $= 9.5\,mm - (-0.75)\,mm = 10.25\,mm$.

Now least count $= GCD(MSD, VSD) = GCD(1mm, 1.25mm) = 0.25\,mm$.

Only correct option is C Ans.

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EXAMPLE :

The main scale of vernier callipers reads in millimeter and its vernier is divided into 8 divisions, which coincide with 6 divisions of main scale. When two jaws of instrument touch each other the zero of the vernier coincide with the zero of main scale. A rod is tightly placed along its length between both jaws, it is observed that the zero of vernier scale lies just right to 45th division of main scale and seventh division of vernier scale coincides with the main scale. Then find the length of the rod in millimeter.

Solution: According to situation of Question, $45\,MSD + x + 7\,VSD = n\,MSD$ where x is a number between 0 and 1mm.

Also, $1\,VSD = 6/8\,MSD = 3/4$ mm.

$$\Rightarrow \quad n = 45 + x + 7 \times \frac{3}{4} = 50.25 + x$$

Now $\because \quad x \in (0,1) \Rightarrow n = 51$

$$\Rightarrow \quad 45\,MSD + x = 51\,MSD - 7\,VSD = 51\text{ mm} - 7 \times \frac{3}{4} = 45.75\text{ mm}$$

$\Rightarrow$ length ≈ 45.75 mm. Ans.

EXAMPLE :

The value of each main scale division of a pair of vernier calipers is 1mm and vernier scale divisions are in A.P. 1st division is 0.95 mm; 2nd division is 0.9 mm; third division is 0.85 mm and so on. When an object is placed between jaws of vernier calipers, zero of vernier lies between 3.1 cm and 3.2 cm and 4th division of vernier coincides with main scale division (neglect the zero error) reading of vernier is (in cm)

Solution: According to situation of Problem:

$$31 + x + 0.95 + 0.9 + 0.85 + 0.8 = n\,MSD = 34.5 + x$$

and $x \in (0,1) \Rightarrow 34.5 + x = n\,MSD \Rightarrow n = 35$

$$\Rightarrow \quad 31 + x = 35\text{ mm} - 3.5\text{ mm} = 31.5\text{ mm} = 3.15\text{ cm}$$ Ans.

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